

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4379

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Unique Paper Code : 32341502

Name of the Paper : Theory of Computation

Name of the Course : B.Sc. (H) Computer Science

Semester : V (Admissions 2019-2021)

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Question No. 1 is compulsory.
3. Attempt any **four** of Question nos. 2 to 7.
4. Parts of a Question must be answered together.
5. Consider $\Sigma = \{a, b\}$ for all the questions unless specified otherwise.

P.T.O.

1. (a) Let $S = \{ab, bb\}$ and $T = \{ab, bb, bbbb\}$.

• Is $S^* \subset T^*$?

• Is $S^* = T^*$?

Explain.

(3)

- (b) Write a regular expression for the language, having words with exactly one double letter in them.

(3)

- (c) Show that the language $\text{PRIME} = \{a^p, \text{ where } p \text{ is a prime}\}$ is non-regular.

(3)

- (d) Prove that the complement of a context-free language may not be context-free.

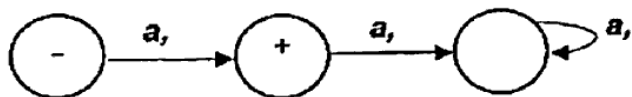
(3)

- (e) (i) What are recursive languages?

(ii) State the Church-Turing thesis.

(3)

- (f) For the following Finite Automata that accepts the language L , draw a deterministic FA that accepts (i) L' (ii) L^* (4)



- (g) Determine whether the following CFG is ambiguous or not. Justify.

$$S \rightarrow aSX \mid \Lambda$$

$$X \rightarrow aX \mid a \quad (4)$$

- (h) Convert the following CFG to Chomsky Normal Form :

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow (E)$$

$$E \rightarrow 7$$

The terminals are $+$, $*$, $(,)$, 7 (4)

P.T.O.

- (i) Design a deterministic PDA for the following language: $a^n b^{n+1}$, $n \geq 1$ (4)
- (j) Design a deterministic finite automata that accepts a string defined over the English alphabet $\{a-z\}$ and ending with 'ied'. For instance, "died" would be accepted, but not "dead". (4)

2. Consider the following languages

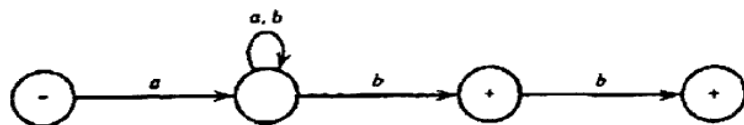
L_{11} : the language of all words that do not contain a double a

L_{12} : the language of all words that do not contain a double b and end in a

- (a) Write a regular expression for each of L_{11} and L_{12} . Hence write a regular expression for $L_{11} \cup L_{12}$. (4)

- (b) Draw a finite automata for each of L_{11} and L_{12} . Hence, systematically find a finite automata for $L_{11} \cap L_{12}$. (6)

3. (a) Convert the following non-deterministic finite automata to deterministic finite automata : (4)



- (b) Design a Turing Machine that decides the language $L = \{a^n b^n c^{n+2} : n \geq 0\}$. (6)

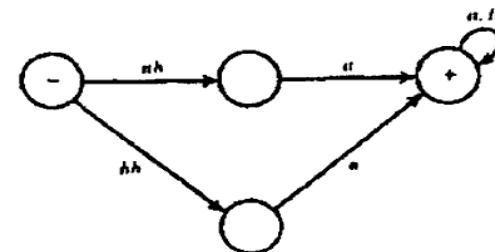
4. (a) Find a CFG for the language $L_4 = a(bb)^*$. Also, find a CFG for L_4^* . (4)

- (b) Construct a deterministic PDA for the language L given below :

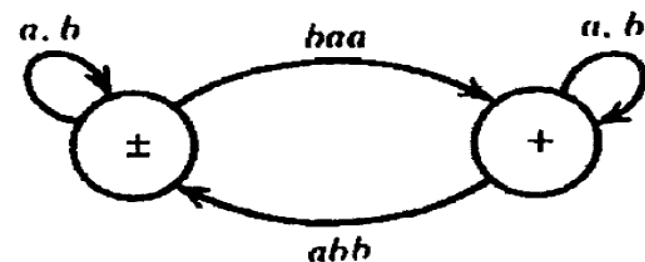
$$L = \{a^m b^n \mid n \geq 1, m \geq 1, m > n+2\} \quad (6)$$

P.T.O.

5. (a) (i) Convert the following Transition graph into a regular expression : (3)



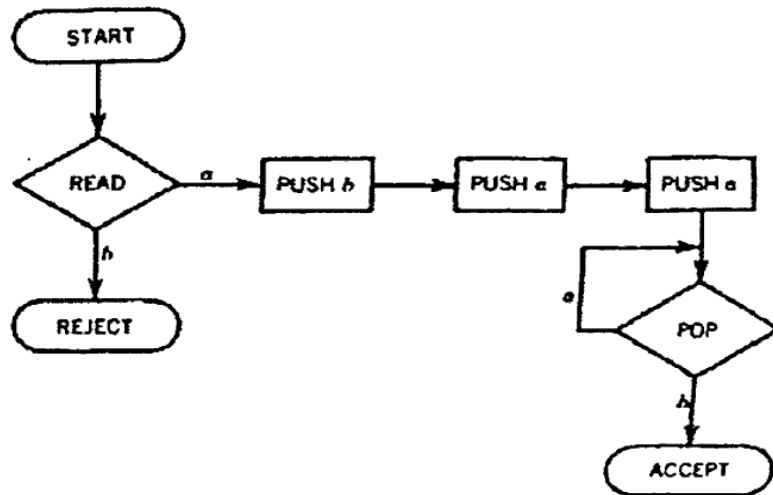
- (ii) Construct a TG for L^* if L is defined by the following TG: <https://www.dudelhi.com> (3)



- (b) Show that the following language defined over the alphabet $\{a, b\}$ is not context free :

$$a^n b^n a^n b^n \quad (4)$$

6. (a) Build a finite automata that accepts the language of words having a double letter in them. (4)
- (b) Prove that recursively enumerable languages are not closed under complementation. (4)
- (c) What would be the length of the shortest string in the language mentioned in part (a) above? List the shortest strings. (2)
7. (a) Describe the language accepted by the following PDA and give its regular expression. (4)



P.T.O.

- (b) Consider the following CFG :

$S \rightarrow aAS$, $S \rightarrow a$, $A \rightarrow SbA$, $A \rightarrow SS$, $A \rightarrow ba$

For the input string 'aabbbaa', find

- (i) leftmost derivation
- (ii) parse tree (6)